

Executive Summary

Revenue & Customer Churn Analytics System

Business Context

Customer churn is one of the most critical challenges for subscription-based businesses, directly impacting recurring revenue and long-term growth. This project was developed to simulate a **real-world churn analytics initiative**, focusing on identifying churn drivers, quantifying revenue risk, and enabling data-backed retention strategies.

The analysis leverages customer-level subscription data and applies industry-standard KPIs to generate insights that would typically be consumed by **product leaders, revenue teams, and executive stakeholders**.

Scope of Analysis

- Customer churn behavior across tenure, contract type, and revenue segments
- Revenue exposure caused by churn-prone customers
- Customer lifetime value distribution
- Early-tenure cohort retention analysis

Key Findings

1. Early-Tenure Churn is the Primary Risk

Customers within their first 6 months show the highest churn rates, indicating onboarding and early engagement as critical intervention points.

2. Contract Structure Strongly Influences Retention

Month-to-month contracts exhibit significantly higher churn compared to long-term contracts, even when controlling for revenue levels.

3. Revenue at Risk is Highly Concentrated

A relatively small subset of customers contributes disproportionately to total revenue at risk, enabling targeted retention strategies rather than broad incentives.

4. Tenure is a Stronger Predictor than Price

While higher monthly charges correlate with churn, tenure and contract length emerge as more reliable predictors.

Business Impact

- Enables **early churn detection** for new customers
- Supports **targeted retention offers** instead of blanket discounts
- Improves revenue forecasting accuracy by quantifying churn-related risk
- Provides executives with **clear, KPI-driven decision support**

Strategic Recommendations (High-Level)

- Strengthen onboarding for first 90 days
- Incentivize long-term contracts for high-value customers
- Deploy early-warning churn signals based on tenure and engagement

Conclusion

This project demonstrates how applied analytics can directly influence revenue protection strategies. By combining data engineering, analytical modeling, and executive visualization, the system reflects how churn analytics is performed in real business environments.